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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}

output = A | B

print(str(output).replace(" ", ","))
```

OUTPUT

{1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

output = A & B

print(str(output).replace(" ", ","))
```

OUTPUT

{2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

output = A - B

print(str(output).replace(" ", ","))
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

output = A ^ B

print(str(output).replace(" ", ","))
```

OUTPUT

{1,4}

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Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not a subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not a superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
else:
    print("Not disjoint")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [1, 2, 2, 3, 3, 4]

output = set(lst)

print(str(output).replace(" ", ","))
```

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

output = A & B & C

print(str(output).replace(", ", ","))
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 5.00 TC: 1/1 passed

CODE

```
Given = {1, 2, 4, 5, 7, 10}

all_numbers = set(range(1, 11))

missing = sorted(all_numbers - Given)

output = "{" + ",".join(map(str, missing)) + "}"

print(output)
```

OUTPUT

{3,6,8,9}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
lst = [10,20,30,20,40]

seen = set()

for num in lst:
    if num in seen:
        print(num)
        break
    seen.add(num)
```

OUTPUT

20

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
list_str = "2 4 6 8 10"
target = 12

lst = list(map(int, list_str.split()))

seen = set()
pairs = []

for num in lst:
    complement = target - num
    if complement in seen:
        pairs.append((complement, num))
    seen.add(num)

output = ",".join(f"({p[0]},{p[1]})" for p in pairs)
print(output)
```

OUTPUT

(4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = "Programming"

unique_chars = set(s)
print(len(unique_chars))
```

OUTPUT

8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
sentence = "python is easy python"

seen = set()
unique_words = []

for word in sentence.split():
    if word not in seen:
        seen.add(word)
        unique_words.append(word)

formatted_items = [f'"{w}"' for w in unique_words]
output = "{" + ",".join(formatted_items) + "}"

print(output)
```

OUTPUT

{'python','is','easy'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
C = {5, 6, 7}

exactly_one = (A ^ B ^ C) - (A & B & C)
sorted_elements = sorted(exactly_one)
output = "{" + ",".join(map(str, sorted_elements)) + "}"

print(output)
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven**CODE**

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
choice = 1

if choice == 1:
    print(f"Union: {set(sorted(A | B))}")
elif choice == 2:
    print(f"Intersection: {set(sorted(A & B))}")
elif choice == 3:
    print(f"Difference: {set(sorted(A - B))}")
elif choice == 4:
    print(f"Symmetric Difference: {set(sorted(A ^ B))}")
elif choice == 5:
    print("Program terminated.")
```

OUTPUT

Union: {1, 2, 3, 4, 5, 6}

Q20. Student Database

CODE

```
import sys

input_lines = sys.stdin.read().splitlines()
students = set()
i = 0

while i < len(input_lines):
    line = input_lines[i].strip()
    i += 1
    if not line:
        continue

    # Handle option input
    if "Enter your choice:" in line:
        choice = line.split(":")[-1].strip()
    else:
        choice = line

    if choice == '1':
        # Add Student
        name = input_lines[i].split(":")[-1].strip()
        i += 1
        students.add(name)
        print("Student added successfully.")

    elif choice == '2':
        # Remove Student
        name = input_lines[i].split(":")[-1].strip()
        i += 1
        if name in students:
            students.remove(name)
            print("Student removed successfully.")
        else:
            print("Student not found.")

    elif choice == '3':
        # Search Student
        name = input_lines[i].split(":")[-1].strip()
        i += 1
        if name in students:
            print("Student found.")
        else:
            print("Student not found.")

    elif choice == '4':
        # Display Students
        print(f"Students: {students}")

    elif choice == '5':
        # Count Students
        print(f"Total number of students: {len(students)}")

    elif choice == '6':
        # Exit
        print("Program terminated.")
        break
```

OUTPUT

Security Error: Disallowed code pattern detected.